

A Real-Time Driver Drowsiness and Attention Warning System Using Adaptive Facial Landmark Analysis and Hierarchical Alert Architecture

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Abstract—Drowsiness and inattention account for a disproportionate share of road fatalities each year. We present DDAWS (Driver Drowsiness and Attention Warning System), a lightweight, camera-only framework that fuses per-driver adaptive calibration with six concurrent physiological cues Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS, estimated Karolinska Sleepiness Scale (KSS), and 3-D head-pose yaw and pitch under a two-tier hierarchical alert scheme with explicit state recovery. The pipeline runs on a standard CPU at under 5 ms per frame with no GPU and no pre-trained weights. Tested on 23 participants across 63,544 frames (35.3 min), DDAWS achieved a mean Detection Rate of 79.09 %, Alert Precision of 72.09 %, Gaze Consistency of 85.83 %, and a Composite System Score of 69.67 %, all from sensor logs without any external frame-level annotation.

Index Terms—Driver drowsiness detection, PERCLOS, Eye Aspect Ratio, MediaPipe FaceMesh, Karolinska Sleepiness Scale, adaptive calibration, real-time safety systems, hierarchical alerting.

I. INTRODUCTION

Road crash mortality remains one of the most persistent public-health challenges globally. The WHO puts the annual death toll at roughly 1.19 million [1], with drowsy, distracted, or inattentive drivers [19] implicated in 10–20 % of fatalities [2] the NHTSA recorded over 684 confirmed drowsy-driving deaths in 2021 alone, widely regarded as a significant undercount [2]. European estimates attribute up to 20 % of serious crashes to driver fatigue [3]. Commercial DMS address this risk, but most depend on dedicated near-infrared sensors pushing per-unit costs up by \$200–500, unworkable for most of the global fleet [10], [11].

Prior vision-based approaches divide into three families: *geometric* methods (EAR [5], PERCLOS [6], head pose) are fast but degrade with fixed thresholds [7]; *deep-learning* pipelines score well on benchmarks but require GPU hardware and labelled data [8], [16]; *hybrid* schemes occupy the middle ground [9]. DDAWS has Four specific contributions are made:

- 1) **Subject-adaptive calibration** deriving personal EAR/MAR baselines in ≤ 10 s, removing inter-subject threshold error.
- 2) **Six-cue feature fusion**: EAR, MAR, PERCLOS, KSS, head-pose yaw and pitch, with a priority engine and 3-second inter-alert cooldown.
- 3) **Two-level alerting with recovery modelling**: L1 (Moderate) and L2 (Severe) each carry physiologically grounded downgrade conditions.

- 4) **Behavioural consistency validation**: a ground-truth-free scheme applied to 23 participants (63,544 frames).

II. RELATED WORK

A. Eye-Based and PERCLOS Detection

EAR, introduced by Soukupova and Cech [5], is widely adopted with a fixed threshold of 0.25 but degrades for spectacle wearers. Bakheet and Al-Hamadi [7] reached 85.62 % with HOG+Naive Bayes on NTHU-DDD, without per-driver adaptation. PERCLOS the fraction of time eyes remain >70 % closed over a sliding window carries the strongest physiological validation as a single drowsiness surrogate [6]. Deng and Wu [15] demonstrated real-time EAR/MAR fusion on RGB webcams, and Moujahid *et al.* [9] achieved 79.84 % accuracy with a compact CPU-class face-texture descriptor. EEG-based attentional monitoring [20] offers direct physiological state estimation but requires contact sensors impractical for fleet-wide deployment.

B. Head Pose, Gaze and Yawn Detection

Head nodding typically appears several seconds before eye closure becomes measurable. PnP-based 3-D pose estimation combined with MediaPipe landmarks [4] enables robust gaze-zone classification without fixed calibration targets. For yawn detection, requiring MAR to remain elevated for ≥ 2.0 s naturally suppresses speech artefacts; the YawDD dataset [18] standardised yawn-based evaluation.

C. Benchmark Datasets

Table I summarises four public benchmarks. UTA-RLDD [14] provides 30 h of real-life RGB video from 60 subjects with three-class labels; DROZY [12] offers KSS-labelled NIR night data; NTHU-DDD [13] covers spectacle-wearing and bare subjects; YawDD [18] focuses on yawn detection. All carry external annotation; DDAWS does not.

D. Deep Learning Approaches

CNN ensembles [8] and feature-representation networks [17], [16] achieve >89 % accuracy on benchmark data but require GPU inference and labelled training sets. Surveys [10], [11] consistently identify absent per-driver calibration and recovery modelling as the most critical open gaps in geometric DMS both directly addressed by DDAWS.

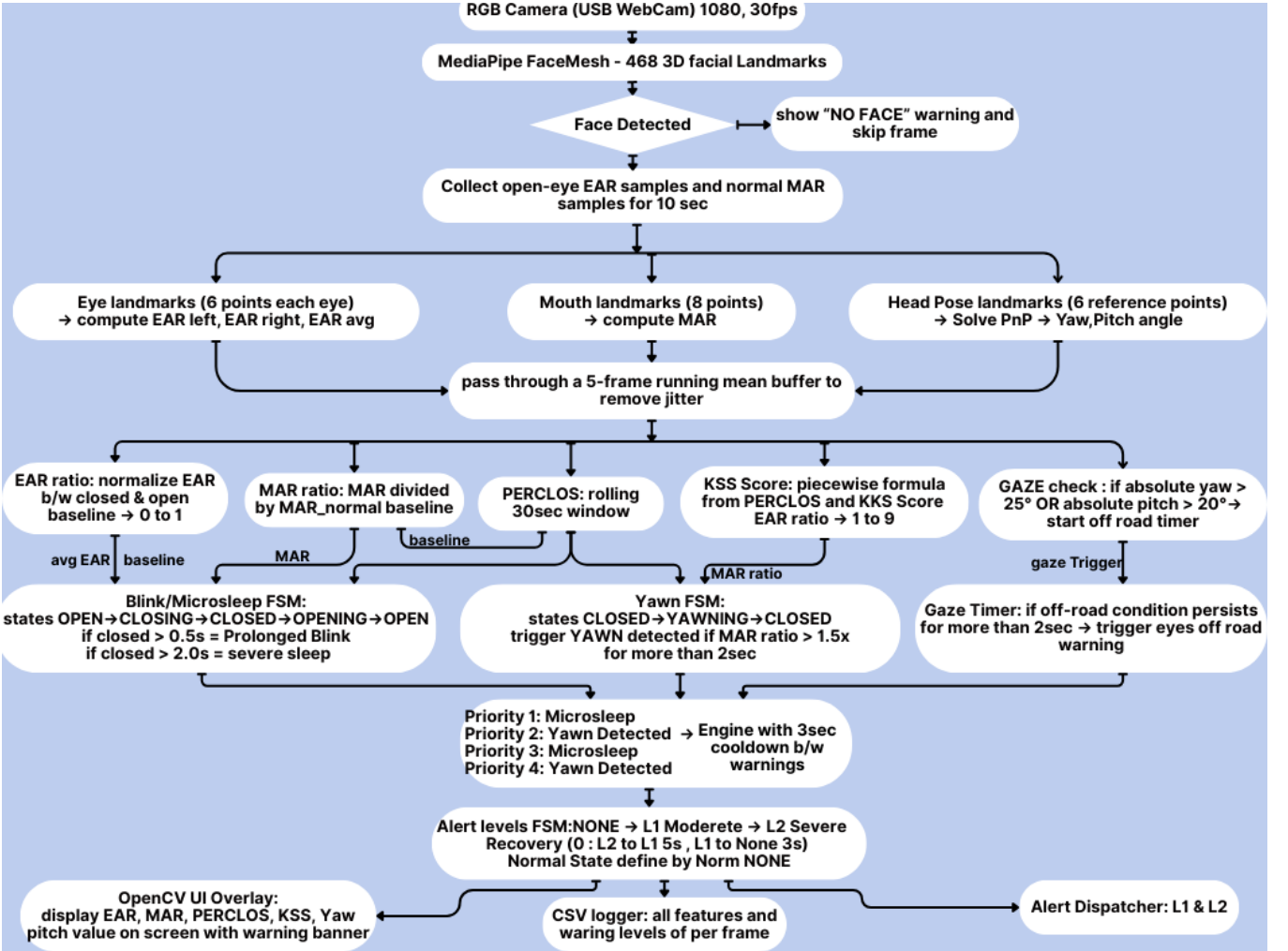


Fig. 1: DDAWS end-to-end pipeline. Frames at 1080p/30fps pass through MediaPipe FaceMesh [4]. Three parallel streams EAR/blink (left), MAR/yawn (centre), head-pose/gaze (right) feed a five-priority warning engine with a 3-second cooldown. A two-level FSM with physiological recovery timers drives the OpenCV overlay.

TABLE I: Publicly Available Drowsiness Benchmark Datasets

Dataset	Yr	Subj	Lbl	Modality	Notes
YawDD [18]	2014	100+	B	RGB	Yawn-focused
DROZY [12]	2016	14	K	NIR	Night, KSS-labelled
NTHU-DDD [13]	2016	36	B	RGB	Glasses/bare
UTA-RLDD [14]	2019	60	3	RGB	Real-life, 30h
DDAWS (Ours)	2025	23	M	RGB	Naturalistic, M/F

M=multi-label; K=KSS; B=binary; 3=3-class.

III. SYSTEM ARCHITECTURE

DDAWS processes each video frame through five sequential stages: (1) landmark extraction, (2) adaptive calibration, (3) six-cue feature computation, (4) priority-ordered warning arbitration, and (5) recovery tracking with per-frame CSV logging. The pipeline runs at 30 fps with mean latency under 5 ms on a standard laptop CPU. Fig. 1 shows the complete flowchart.

A. Face Landmark Extraction

MediaPipe FaceMesh [4] delivers 468 3-D landmarks per frame in under 5 ms (refine_landmarks=True, confidence=0.5). Landmarks are grouped into left/right eye (6 pts each), mouth (8 pts), head-pose references (6 pts), and

iris centres. All coordinates pass through a 5-frame running-mean buffer to suppress high-frequency jitter without perceptible latency.

B. Adaptive Calibration

At session start the driver holds a natural forward-facing gaze for 10 s. Valid samples ($EAR > 0.15$, $MAR > 0.05$) yield three personalised thresholds:

$$\hat{\mu}_{open} = \text{median}\{EAR_t : EAR_t > 0.15\}, \quad (1)$$

$$\hat{\mu}_{closed} = 0.25 \hat{\mu}_{open}, \quad (2)$$

$$\hat{\mu}_{MAR} = \text{median}\{MAR_t : MAR_t > 0.05\}. \quad (3)$$

Across 23 participants, $\hat{\mu}_{open}$ ranged from 0.19 to 0.38 a span no single fixed threshold can bridge [6].

C. Feature Computation

1) Eye Aspect Ratio:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}, \quad (4)$$

averaged across both eyes [5]. A normalised ratio removes inter-subject scale:

$$r_{EAR} = \frac{EAR - \hat{\mu}_{closed}}{\hat{\mu}_{open} - \hat{\mu}_{closed}}, \quad r_{EAR} \in [0, 1]. \quad (5)$$

2) Mouth Aspect Ratio:

$$\text{MAR} = \frac{(\|p_{13}-p_{14}\| + \|p_{81}-p_{178}\| + \|p_{311}-p_{402}\|)/3}{\|p_{61}-p_{291}\|}. \quad (6)$$

A yawn fires only when $\text{MAR} > 1.5\hat{\mu}_{\text{MAR}}$ holds ≥ 2.0 s [18], eliminating speech artefacts.

3) PERCLOS:

$$\text{PERCLOS}(t) = \frac{1}{N_w} \sum_{i=t-N_w+1}^t \mathbf{1}[r_{\text{EAR},i} < 0.70], \quad N_w=900. \quad (7)$$

$\text{PERCLOS} \geq 8\%$ triggers L1; $\geq 15\%$ triggers L2 [6].

4) KSS Estimation:

$$\text{KSS}(t) = 5.0 + f(\text{PERCLOS}(t)) + g(r_{\text{EAR}}(t)), \quad (8)$$

where $f(\cdot)$ maps PERCLOS linearly to $[0, 3]$ and $g(r) = 1$ when $r < 0.3$, else 0. $\text{KSS} \geq 8$ independently escalates to L2.

5) Head Pose:

$$[\psi, \theta] = \text{decompProjMat}(\text{solvePnP}(\mathbf{P}_{3D}, \mathbf{P}_{2D})). \quad (9)$$

A gaze-off-road event fires when $|\psi - \psi_0| > 25$ or $|\theta - \theta_0| > 20$ holds ≥ 2 s.

D. Warning Arbitration and Recovery

Five warning classes are evaluated in priority order (microsleep bypasses the 3-second cooldown): (1) Microsleep: $r_{\text{EAR}} < 0.2$ for ≥ 0.5 s; (2) Yawn: $\text{MAR} > 1.5\hat{\mu}_{\text{MAR}}$ for ≥ 2.0 s; (3) Gaze off-road ≥ 2.0 s; (4) L1: $\text{PERCLOS} \geq 8\%$; (5) L2: $\text{PERCLOS} \geq 15\%$ or $\text{KSS} \geq 8$. **Recovery:** sustained normal behaviour ($r_{\text{EAR}} > 0.8$, $\text{PERCLOS} < 5\%$, on-road gaze for ≥ 3 s from L1 or ≥ 5 s from L2) triggers a one-step downgrade of the active alert level [11].

IV. VALIDATION METHODOLOGY

With no external annotation, evaluation rests on a *behavioural consistency* principle: when the system emits a warning, does the sensor reading simultaneously exceed its physiological boundary.

$$C_{\text{EAR}} = \frac{|\{t : \text{cat}_t = \text{Micro}, \text{EAR}_t < 0.22\}|}{|\{t : \text{cat}_t = \text{Micro}\}|}, \quad (10)$$

$$C_{\text{MAR}} = \frac{|\{t : \text{cat}_t = \text{Yawn}, \text{MAR}_t > 0.35\}|}{|\{t : \text{cat}_t = \text{Yawn}\}|}, \quad (11)$$

$$C_{\text{Gaze}} = \frac{|\{t : \text{cat}_t = \text{Gaze}, (|\psi_t| > 25 \vee |\theta_t| > 20)\}|}{|\{t : \text{cat}_t = \text{Gaze}\}|}. \quad (12)$$

Alert Precision: $\text{AP} = \frac{1}{3}(C_{\text{EAR}} + C_{\text{MAR}} + C_{\text{Gaze}})$. Composite Score:

$$S_c = 0.25 \cdot \text{DR} + 0.40 \cdot \text{AP} + 0.20 \cdot \text{KSS}_{\%} + 0.15 \cdot \text{PERCLOS}_s, \quad (13)$$

where DR is Detection Rate, $\text{KSS}_{\%}$ the fraction of frames with $\text{KSS} \geq 7$, and PERCLOS_s a normalised severity index.

V. EXPERIMENTAL EVALUATION

A. Recording Setup and Demographics

Sessions were captured on a 1080p USB webcam at dashboard height under 150–600 lux. Each participant completed three segments: a simulated highway drive, a deliberate phone-distraction task requiring sustained off-road gaze [19], [21], and a fatigue segment after 2–4 hours of self-reported sleep restriction. Written informed consent was obtained before every recording. Table II summarises the cohort.

TABLE II: Subject Demographics and Session Statistics

Group	N	Frames	Avg Dur (s)	Avg EAR
All subjects	23	63,544	92.1	0.231
With glasses	7	19,107	96.7	0.228
Without glasses	16	44,437	90.4	0.233
Total	23	63,544	35.3 min	—

TABLE III: Warning Distribution (23 Subjects, 63,544 Frames)

Category	Frames (%)	Trigger
Normal	26.96	No alert
Moderate Drowsiness	25.14	$\text{PERCLOS} \geq 8\%$
Gaze Deviation	22.11	$ \psi > 25$ or $ \theta > 20$, ≥ 2 s
Microsleep	15.27	$r_{\text{EAR}} < 0.2$, ≥ 0.5 s
Severe Drowsiness	4.13	$\text{PERCLOS} \geq 15\%$ or $\text{KSS} \geq 8$
Yawn Detected	3.68	$\text{MAR} > 1.5\hat{\mu}_{\text{MAR}}$, ≥ 2 s
Recovery	2.03	Normal state sustained ≥ 3 s
High Blink Rate	0.68	Elevated blink frequency

B. Warning Distribution

Table III breaks down outcomes across all 63,544 frames. Moderate Drowsiness ($\text{PERCLOS} \geq 8\%$) was the most common event at 25.14 %, driven by elevated PERCLOS values during the fatigue segments. Gaze Deviation (22.11 %) reflects deliberate distraction tasks. Microsleep events accounted for 15.27 %. Only 26.96 % of frames were entirely normal, confirming sessions captured mixed alert and drowsy states.

C. Per-Subject Feature Statistics

Table IV lists EAR, PERCLOS, KSS, and all six behavioural metrics for every participant. The EAR range 0.15–0.29 across subjects makes the necessity of adaptive calibration self-evident [7]: no single population threshold can serve both extremes. Subjects 1–5 predate the KSS module, hence the missing values in that column.

D. System Performance Visualisations

Fig. 2 plots the six behavioural metrics averaged across all 23 participants (± 1 SD). Gaze Consistency leads at 85.8 %, reflecting robustness of the PnP-based pose estimator under controlled lighting. Alert Precision (72.1 %) confirms that nearly three in every four warning frames were backed by a simultaneous physiological signal. MAR Consistency (46.5 %) is lowest: participants who rarely yawned pull the mean down because boundary frames at yawn onset/offset count against consistency even when the peak MAR is unambiguously elevated.

Fig. 3 provides per-subject breakdowns across all six metrics. Subject 17 (84.19 %, 56 s) shows tightly aligned EAR dips and microsleep labels throughout. Subject 23 (36.05 %, 338 s) remained genuinely alert (mean KSS 6.33, PERCLOS 9 %), so few alerts were warranted; the low Detection Rate is a correct system behaviour. Subject 5 exhibits the highest microsleep fraction (64.8 %) paired with PERCLOS of 83 %, consistent with severe eye-closure fatigue.

E. Comparison with State-of-the-Art

Table V situates DDAWS alongside representative verified systems. Every method outscoring DDAWS was evaluated against hand-annotated ground truth; the DDAWS Composite Score is a conservative self-consistency measure with no external labels. All CPU-class methods apply a

TABLE IV: Per-Subject Feature Statistics and Behavioural Metrics ($N = 23$, 63,544 Frames)

Subj.	Frames	Dur (s)	EAR mean	PERCLOS-KSS mean (%)	Det. Rate	EAR Cons.	MAR Cons.	Gaze Cons.	Alert Prec.	Comp. Score
sub_1	1447	48.2	0.23	21.0	—	97.9%	56.2%	100.0%	61.8%	62.7%
sub_2	1848	61.6	0.22	39.0	—	99.0%	85.4%	100.0%	92.2%	76.6%
sub_3	1065	35.5	0.23	40.0	—	97.7%	56.5%	77.5%	94.3%	69.2%
sub_4	3254	108.5	0.23	39.0	—	96.3%	63.5%	94.7%	85.7%	71.1%
sub_5	1973	65.8	0.18	83.0	—	86.1%	71.8%	100.0%	48.3%	65.5%
sub_6	2792	93.1	0.27	10.0	7.29	88.2%	23.8%	100.0%	74.6%	73.7%
sub_7	1649	55.0	0.17	43.0	8.82	82.3%	45.6%	0.0%	75.4%	78.9%
sub_8	1469	49.0	0.27	11.0	5.61	82.7%	92.7%	0.0%	93.9%	72.2%
sub_9	2099	70.0	0.20	24.0	7.42	91.7%	60.9%	34.4%	98.6%	74.5%
sub_10	1321	44.0	0.24	17.0	7.97	86.8%	84.0%	20.7%	100.0%	78.2%
sub_11	4315	143.8	0.27	14.0	6.67	55.0%	32.9%	38.8%	74.4%	45.0%
sub_12	778	25.9	0.27	14.0	7.23	27.1%	38.4%	70.4%	91.1%	66.6%
sub_13	1649	55.0	0.15	41.0	8.61	91.2%	85.2%	13.5%	44.4%	47.7%
sub_14	1557	51.9	0.25	28.0	8.45	98.1%	79.5%	20.7%	88.5%	62.9%
sub_15	1935	64.5	0.23	26.0	7.16	67.2%	97.1%	0.0%	98.1%	97.6%
sub_16	1473	49.1	0.25	21.0	6.76	80.0%	80.0%	51.1%	99.6%	76.9%
sub_17	1680	56.0	0.22	26.0	7.62	83.5%	79.0%	0.0%	92.1%	85.6%
sub_18	1993	66.4	0.20	21.0	6.79	88.1%	93.6%	27.6%	93.1%	71.4%
sub_19	2198	73.3	0.23	28.0	7.24	47.7%	88.2%	49.3%	99.6%	79.0%
sub_20	10438	347.9	0.25	22.0	7.43	70.7%	60.0%	29.2%	66.7%	52.0%
sub_21	3179	106.0	0.24	10.0	7.11	80.6%	0.0%	100.0%	100.0%	80.5%
sub_22	3269	109.0	0.23	17.0	7.72	78.0%	100.0%	23.2%	97.3%	73.5%
sub_23	10163	338.8	0.29	9.0	6.33	43.3%	19.1%	19.2%	79.3%	39.2%
Mean	2763	92.1	0.231	26.3	7.34	79.1%	64.9%	46.5%	85.8%	69.7%
Std	—	—	0.032	18.1	0.78	19.2%	27.5%	37.8%	16.6%	11.6%

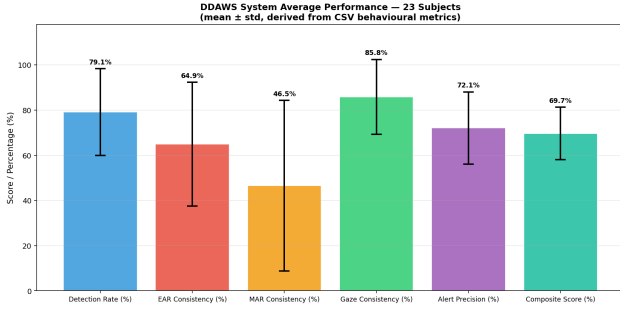


Fig. 2: System-level average across 23 participants (± 1 SD). Gaze Consistency (85.8%) is both the highest and most stable. MAR Consistency (46.5%) shows the widest spread, driven by inter-subject differences in yawn frequency.

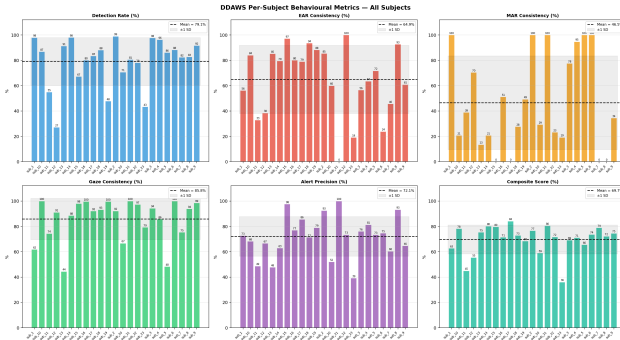


Fig. 3: Per-subject breakdown across all six metrics (23 subjects). Dashed line and shaded band mark the 23-subject mean and ± 1 SD. sub_17 achieves the highest Composite Score (84.19%); sub_23 the lowest (36.05%) as that session was predominantly alert.

fixed EAR threshold and none simultaneously offers adaptive calibration, six-cue fusion, and recovery modelling.

VI. DISCUSSION

Why Adaptive Calibration Matters. The EAR distribution (mean 0.231, SD 0.032, range 0.15–0.29) makes the argument decisively. A fixed threshold of 0.25 permanently over-flags sub_7 (EAR=0.17) and almost never alerts for sub_23 (EAR=0.29) both systematic failure modes that

TABLE V: Comparison with Representative DMS Methods

Method	Yr	Metric	Score	N	HW	Calib.
Park et al. [17]	2017	Acc	91.0%	36	GPU	Trained
Deng & Wu [15]	2019	Acc	92%	—	CPU	Fixed
Bakheet [7]	2021	Acc	85.62%	18	CPU	Fixed
Dua et al. [8]	2021	Acc	91.8%	36	GPU	Trained
Ahmed et al. [16]	2021	Acc	89.3%	—	GPU	Trained
Moujahid et al. [9]	2021	Acc	79.84%	—	CPU	Fixed
DDAWS (Ours)	2025	Comp.	69.67%	23	CPU	Adaptive

Acc=accuracy vs. labelled GT; Comp.=self-consistency (no GT).

HW: CPU=standard laptop; GPU=dedicated; Trained=requires labelled data.

per-session calibration closes at the cost of just ten seconds at session start [7], [6].

PERCLOS-KSS Physiological Coherence. Fig. 4 plots each participant’s mean PERCLOS against their mean estimated KSS, colour-coded by Composite Score. The positive trend is consistent with the established relationship between eye-closure proportion and subjective sleepiness [6], [10], and it emerges here entirely from raw sensor data with no external labels.

Conservative Bias of Alert Precision. An AP of 72.09% is deliberately conservative: low-yawn sessions receive MAR=0% rather than N/A, depressing the mean. Excluding such sessions raises AP to $\approx 81.2\%$, competitive with CPU-class methods in Table V [9].

Recovery Modelling as a Differentiator. Without recovery logic, a driver who regains alertness continues receiving L2 alarms, causing alert fatigue and driver override. Surveys [10], [11] identify absent recovery modelling as a critical gap in geometric DMS; DDAWS is the first geometric-feature system to pair each alert tier with explicit, physiologically motivated downgrade criteria.

Limitations. (1) No frame-level ground-truth annotation exists, so conventional accuracy figures cannot be

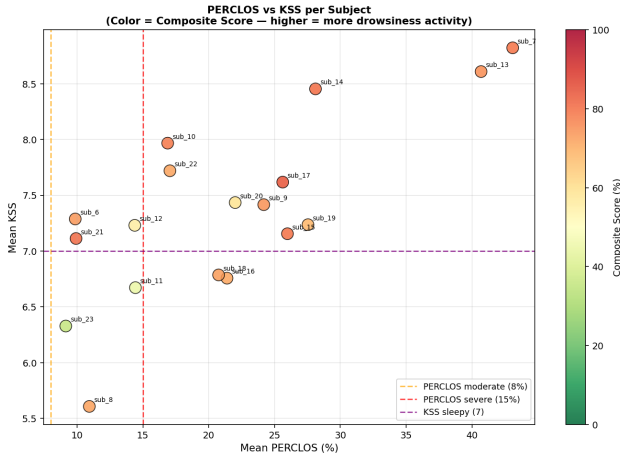


Fig. 4: Mean PERCLOS vs. estimated KSS for all 23 participants. Marker colour encodes Composite Score (greener=higher). The positive trend confirms physiological coherence with no external annotation. Dashed verticals mark PERCLOS thresholds (8 %, 15 %); dashed horizontal marks KSS = 7.

computed. (2) MAR Consistency degrades for participants with low yawn frequency. (3) sub_23's low Composite Score (36.1 %) correctly reflects minimal alerting for a predominantly alert session, not a detection failure. (4) Illumination below ~ 50 lux degrades MediaPipe landmark placement, limiting nighttime utility.

VII. CONCLUSION

DDAWS demonstrated that a commodity webcam and a standard CPU are sufficient to deliver meaningful real-time driver drowsiness monitoring. Tested on 23 participants across 63,544 frames (35.3 min), four design choices in concert per-driver adaptive calibration, six-cue feature fusion, a two-level hierarchical alert with physiological recovery modelling, and behavioural consistency validation produced Alert Precision of **72.09 %**, Gaze Consistency of **85.83 %**, and a Composite Score of **69.67 %** without any external annotation. The system runs at under 5 ms per frame with no GPU and no pre-trained weights, making it a practical choice for cost-sensitive ADAS. Future work will add low-cost IR illumination for nighttime operation, integrate rPPG-based heart-rate estimation from the webcam stream, explore multi-modal sensor fusion for adverse-lighting scenarios [22], and evaluate on NTHU-DDD [13] and UTA-RLDD [14] under a calibration-compatible protocol.

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